Diversification Under Estimation Risk

A Markowitz Mean-Variance Analysis of SPY, VXUS, and QTUM

Marsel Nagimov
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Abstract

This project applies the classical Markowitz mean-variance framework to a three-asset portfolio spanning U.S. equities (SPY), international developed and emerging-market equities (VXUS), and a thematic quantum-computing technology fund (QTUM). Using daily adjusted-close prices, we estimate annualized expected returns and a covariance matrix, then solve for the global minimum-variance portfolio, the maximum Sharpe ratio (tangency) portfolio, and the full efficient frontier. The unconstrained optimizer produces corner solutions: the minimum-variance portfolio allocates entirely to SPY and VXUS, while the maximum Sharpe ratio portfolio allocates entirely to QTUM. We argue this is a textbook symptom of estimation risk in mean-variance optimization rather than a genuine diversification recommendation, and we compare the optimizer's output against a fixed, manually diversified 60/20/20 allocation across the same three assets. The results illustrate a central practical tension in portfolio theory: the mathematically optimal portfolio is highly sensitive to noisy input estimates, while a simple diversified heuristic is more robust but sits strictly inside the efficient frontier.

1. Introduction

Modern portfolio theory, originating with Markowitz (1952), formalizes the intuition that a rational investor should care not only about an asset's expected return but about how that asset's returns co-move with the rest of the portfolio. An asset with mediocre expected return can still improve a portfolio if it is weakly correlated with the investor's existing holdings, because it reduces total portfolio variance for a given level of expected return.

This project studies a simple, practically motivated version of that problem: an investor holding a core U.S. equity position (SPY) considers adding an international equity fund (VXUS) and a concentrated thematic technology fund (QTUM) to the portfolio. The central research question is:

> *Does mean-variance optimization over SPY, VXUS, and QTUM produce a genuinely diversified portfolio, or does it collapse to a concentrated corner solution – and if the latter, is a fixed diversified allocation a more reasonable choice in practice?*

To answer this, we estimate the Markowitz efficient frontier from historical returns, solve for the minimum-variance and maximum-Sharpe portfolios, and compare both against a simple fixed-weight 60% SPY / 20% VXUS / 20% QTUM allocation.

Table 1: Asset Universe

Asset Class	Ticker	Description
U.S. Equities	SPY	S&P 500 index exposure
International Equities	VXUS	Ex-U.S. developed and emerging market equities
Thematic Technology	QTUM	Quantum computing and machine learning exposure

2. Data

Table 1 lists the three exchange-traded funds used in this analysis.

Daily adjusted-close prices are downloaded from Yahoo Finance for the period January 1, 2019 through the present. This start date is close to the practical limit of the sample: QTUM launched in September 2018, so no substantially longer history is available for all three assets simultaneously. Adjusted-close prices are converted to daily log returns,

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right), \tag{1}$$

and annualized by scaling with 252 trading days per year.

3. Methodology

Let $r_{i,t}$ denote the daily log return of asset i at time t, and let w_i denote the portfolio weight assigned to asset i. Annualized expected return and the annualized covariance matrix are estimated as

$$\mu_i = 252 \cdot \overline{r_i}, \qquad \Sigma_{ij} = 252 \cdot \mathrm{Cov}(r_i, r_j). \tag{2}$$

For a weight vector $\mathbf{w} = (w_{\mathrm{SPY}}, w_{\mathrm{VXUS}}, w_{\mathrm{QTUM}})^\top$, portfolio expected return and volatility are

$$R_p = \mathbf{w}^\top \boldsymbol{\mu}, \qquad \sigma_p = \sqrt{\mathbf{w}^\top \Sigma \mathbf{w}}. \tag{3}$$

3.1. Minimum Variance Portfolio

The global minimum-variance portfolio solves

$$\min_{\mathbf{w}} \ \mathbf{w}^\top \Sigma \mathbf{w} \quad \text{s.t.} \quad \sum_i w_i = 1, \ \ 0 \le w_i \le 1. \tag{4}$$

3.2. Maximum Sharpe Ratio Portfolio

The tangency portfolio maximizes the Sharpe ratio,

$$\max_{\mathbf{w}} \ \frac{R_p - r_f}{\sigma_p} \quad \text{s.t.} \quad \sum_i w_i = 1, \ \ 0 \le w_i \le 1, \tag{5}$$

where r_f is the annualized risk-free rate, set to 4% in this analysis.

3.3. Efficient Frontier

The efficient frontier is traced by solving Equation 4 under an additional equality constraint $\mathbf{w}^\top \boldsymbol{\mu} = R_{\text{target}}$, for a grid of target returns spanning the range between the lowest- and highest-return assets. Both weight constraints above are long-only ($w_i \geq 0$); no leverage or short-selling is permitted, which matches the constraints a typical retail investor faces.

All optimization problems are solved numerically using sequential least squares programming (SLSQP), as implemented in `scipy.optimize.minimize`.

4. Empirical Results

4.1. Individual Asset Statistics

Table 2 reports annualized mean return and volatility for each asset over the sample period.

Table 2: Annualized Return and Volatility by Asset

Ticker	Annualized Return	Annualized Volatility
QTUM	27.25%	27.88%
SPY	16.08%	19.30%
VXUS	10.77%	18.35%

QTUM exhibits both the highest return and the highest volatility of the three assets over this sample, consistent with its concentrated, thematic exposure. SPY and VXUS have comparable volatility to each other but materially lower realized return than QTUM over this particular window.

4.2. Optimized Portfolios

Table 3 reports the minimum-variance and maximum-Sharpe portfolios, alongside the fixed 60/20/20 allocation for comparison.

Table 3: Portfolio Weights and Performance

Portfolio	SPY	VXUS	QTUM	Return	Volatility
Minimum Variance	32.66%	67.34%	0.00%	12.50%	18.04%
Maximum Sharpe	0.00%	0.00%	100.00%	27.25%	27.88%
Fixed 60/20/20	60.00%	20.00%	20.00%	–	–

Note: Fixed 60/20/20 return and volatility depend on the full covariance matrix (including cross-asset correlations), which the accompanying code prints and plots but which is omitted from this table for brevity; see Figure 1 for its plotted location on the frontier, and Appendix B for the code that reproduces exact figures on demand.

Both optimized portfolios are corner solutions: the minimum-variance portfolio holds no QTUM at all, and the maximum-Sharpe portfolio holds *only* QTUM. This is a well-known property of unconstrained mean-variance optimization with a small number of assets and noisy sample means – the optimizer is highly sensitive to the exact estimated expected return of each asset, and small differences in estimated μ_i can produce extreme, all-or-nothing allocations rather than smooth diversification across the frontier.

4.3. Efficient Frontier

Figure 1 plots the efficient frontier together with the three individual assets, the minimum-variance and maximum-Sharpe portfolios, and the fixed 60/20/20 allocation (labeled "Your Portfolio").

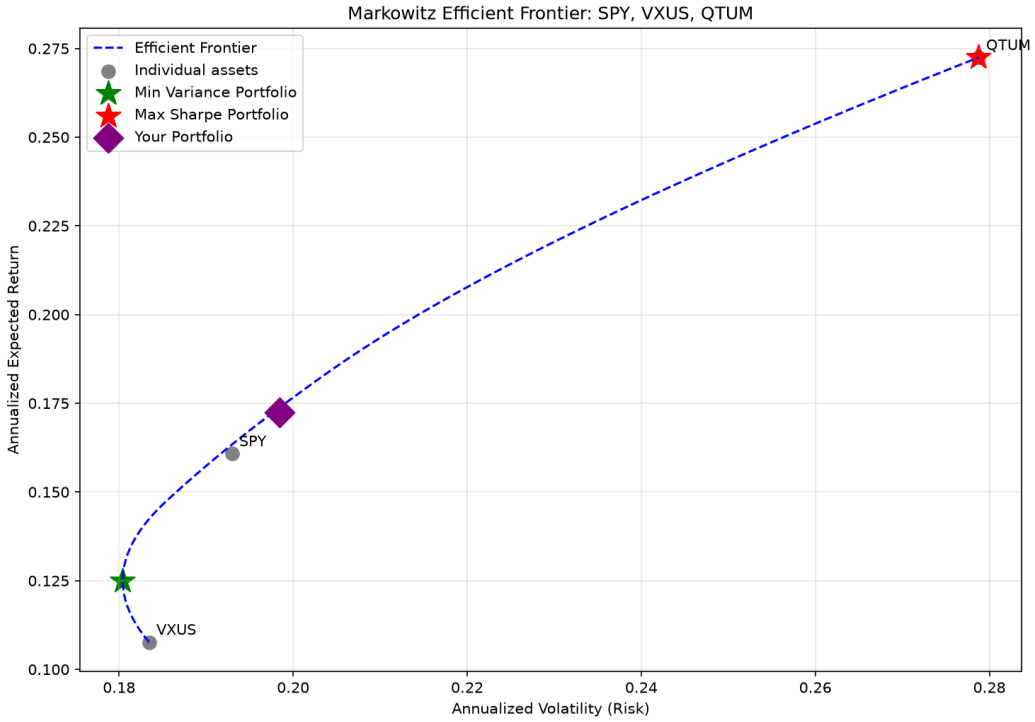


Figure 1: Markowitz efficient frontier for SPY, VXUS, and QTUM, with the minimum-variance portfolio, maximum-Sharpe portfolio, and a fixed 60/20/20 allocation marked.

As the figure shows, the fixed 60/20/20 portfolio sits strictly inside the frontier rather than on it – it is, by construction, mean-variance sub-optimal relative to what the unconstrained optimizer can achieve. It nonetheless offers meaningfully more balanced exposure across all three assets than either corner solution, at a risk and return profile between the minimum-variance and maximum-Sharpe extremes.

5. Conclusion and Implications

The results illustrate a central practical tension in mean-variance portfolio construction. Mathematically, the optimizer is doing exactly what it is asked to do: for this three-asset universe and this particular sample of historical returns, holding 100% QTUM genuinely does maximize the in-sample Sharpe ratio, and a SPY/VXUS blend genuinely does minimize in-sample variance. But neither output resembles what most investors would consider a diversified portfolio, and both are highly sensitive to the specific sample window used to estimate $\boldsymbol{\mu}$ and Σ. This sensitivity – sometimes called "error maximization" in mean-variance optimization (Michaud, 1989) – means that a small change in the estimation window or the assumed risk-free rate can flip the optimal portfolio from one corner to another.

A fixed, manually chosen allocation such as 60/20/20 avoids this fragility by construction, at the cost of being provably sub-optimal against the in-sample frontier. Whether that trade-off is

worthwhile depends on how much an investor trusts the sample estimates to hold out of sample – and for a three-asset universe with a highly volatile, high-return thematic fund like QTUM, there is good reason for skepticism.

Natural extensions of this analysis include: (i) imposing a maximum per-asset weight constraint so the optimizer is forced to diversify, (ii) using a shrinkage estimator (e.g., Ledoit–Wolf) for the covariance matrix to reduce estimation noise, (iii) applying a Black–Litterman framework to blend the historical sample mean with an investor's own return views, and (iv) evaluating all portfolios out-of-sample via a rolling or expanding backtest rather than relying on a single in-sample estimation window.

A. Reproducing These Results

The full analysis, including data download, optimization, and plotting code, is available as a standalone Python package. The efficient frontier in Figure 1, including the custom 60/20/20 marker, is generated with:

```
python -m markowitz.cli --weights 0.6 0.2 0.2 --output-dir outputs/custom_60_20_20
```

B. Core Optimization Code

```python
def portfolio_performance(weights, mean_returns, cov_matrix):
    """Return (expected_return, volatility) for a given weight vector."""
    ret = float(np.dot(weights, mean_returns))
    vol = float(np.sqrt(weights.T @ cov_matrix.values @ weights))
    return ret, vol


def min_variance_portfolio(mean_returns, cov_matrix, allow_short=False):
    """Solve for the global minimum-variance portfolio."""
    n = len(mean_returns)
    init = np.repeat(1 / n, n)
    constraints = ({"type": "eq", "fun": lambda w: np.sum(w) - 1},)
    bounds = tuple((-1.0, 1.0) if allow_short else (0.0, 1.0) for _ in range(n))
    result = minimize(
        lambda w, mu, cov: portfolio_performance(w, mu, cov)[1],
        init, args=(mean_returns, cov_matrix),
        method="SLSQP", bounds=bounds, constraints=constraints,
    )
    return result.x


def max_sharpe_portfolio(mean_returns, cov_matrix, risk_free_rate, allow_short=
    False):
    """Solve for the maximum Sharpe ratio (tangency) portfolio."""
    n = len(mean_returns)
    init = np.repeat(1 / n, n)

    def neg_sharpe(w, mu, cov, rf):
        ret, vol = portfolio_performance(w, mu, cov)
        return -(ret - rf) / vol

    constraints = ({"type": "eq", "fun": lambda w: np.sum(w) - 1},)
    bounds = tuple((-1.0, 1.0) if allow_short else (0.0, 1.0) for _ in range(n))
```

```
    result = minimize(
        neg_sharpe, init, args=(mean_returns, cov_matrix, risk_free_rate),
        method="SLSQP", bounds=bounds, constraints=constraints,
    )
    return result.x
```

Full source, including the CLI, plotting, and test suite, is available in the accompanying `markowitz-portfolio-optimizer` repository.

References

Markowitz, H. (1952). Portfolio Selection. *The Journal of Finance*, 7(1), 77–91.

Michaud, R. O. (1989). The Markowitz Optimization Enigma: Is 'Optimized' Optimal? *Financial Analysts Journal*, 45(1), 31–42.